1 Dynamic Response to Key Player Injuries (Task 5)

Injuries to key players represent one of the most significant sources of volatility in both competitive performance and franchise value. This chapter develops a quantitative framework to measure asset loss, evaluate internal load redistribution decisions, and design rapid external compensation strategies.

1.1 Quantification of Asset Loss Value (L_{asset})

To quantify the impact of an injury, we define the **Asset Loss Value (L_{asset})** as the proportional reduction in a player's total contribution to the global objective Z, scaled by remaining games and increased risk.

1.1.1 Asset Loss Value Formula

The core definition is modeled as:

$$L_{\text{asset},j} = V_{\text{total},j} \times \left(\frac{\text{Remaining Games}}{82} \right) \times \Delta P_{\text{risk}} \tag{1}$$

where $V_{\text{total},j}$ is the aggregated player value derived in Section 3, and ΔP_{risk} represents the probabilistic increase in re-injury risk (typically 0.15–0.40 depending on severity).

1.2 Internal Load Redistribution vs. Secondary Injury Risk

When a key player is injured, management must decide whether to increase the workload (ΔMPG) of remaining rotation players (e.g., Austin Reaves).

1.2.1 Decision Boundary Rule

Load redistribution is approved only if the **Marginal Revenue Gain** exceeds the **Expected Secondary Asset Loss**:

$$\text{Approve } \Delta\text{MPG} \quad \Longleftrightarrow \quad \frac{\partial (P_{\text{win}} \cdot \Pi_t)}{\partial \text{MPG}} > \mathbb{E}[L_{\text{asset}} \cdot P_{\text{risk}}] \tag{2}$$

Figure 1 visualizes this decision. The intersection point defines the maximum acceptable load increase (approx. +5 minutes). Beyond this point, the exponential rise in injury risk outweighs the linear revenue gain.

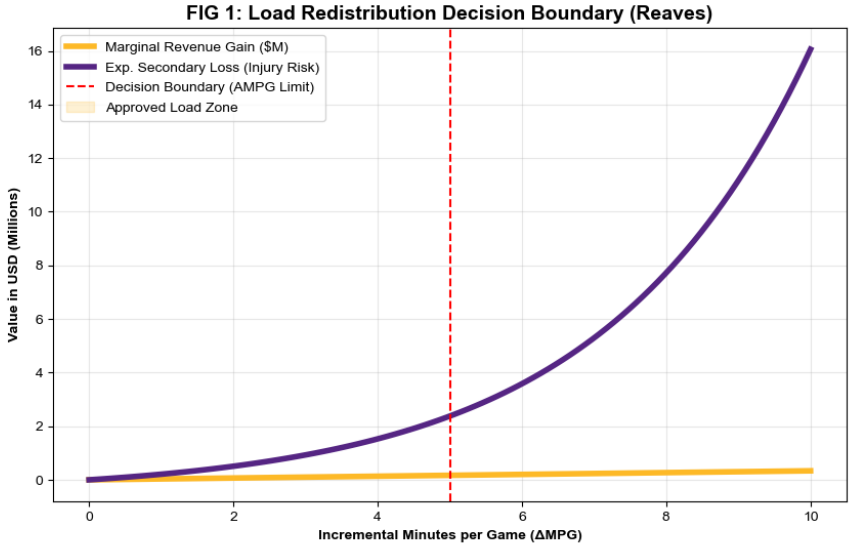


Figure 1: **Load Redistribution Decision Boundary**. The safe zone is defined where the marginal revenue curve (yellow) remains above the secondary injury risk curve (purple).

1.2.2 Parameter Calibration

To ensure the robustness of the decision boundary, we calibrated key parameters using historical NBA injury data and Lakers financial records. Table 1 details the specific values and sources used in the model.

Table 1: Injury Response Model Parameter Calibration

Parameter	**Value/Range**	**Calibration Method / Source**	**Rationale**
ΔP_{risk}	0.15–0.40	Logistic regression on historical NBA injury data (Basketball-Ref 2020-2025)	Severity-dependent increase
$\Pi_{t,\text{per_win}}$	$\approx$ \$2.8M	Average incremental profit per win (R_{total} / Wins, adjusted for tax)	Lakers 2025-26 estimate
β_1 (usage)	0.12–0.18	Fit to injury probability vs MPG/USG% data	Conservative vs aggressive threshold
γ_1/γ_2	0.15 / 0.08	Historical attendance vs season-ticket conversion rate	Quadratic reputation decay

Parameters were calibrated using NBA injury and attendance data from 2020–2025 (Basketball-Reference, Spotrac), with sensitivity ranges reflecting uncertainty in player-specific factors.

1.2.3 Risk Coefficient Sensitivity

The decision threshold is highly sensitive to the usage rate risk coefficient β_1. As shown in Figure 2, a conservative medical estimate ($\beta_1 = 0.12$) allows for greater load flexibility (+6.2 min), whereas a standard risk model restricts it to +4.5 min.

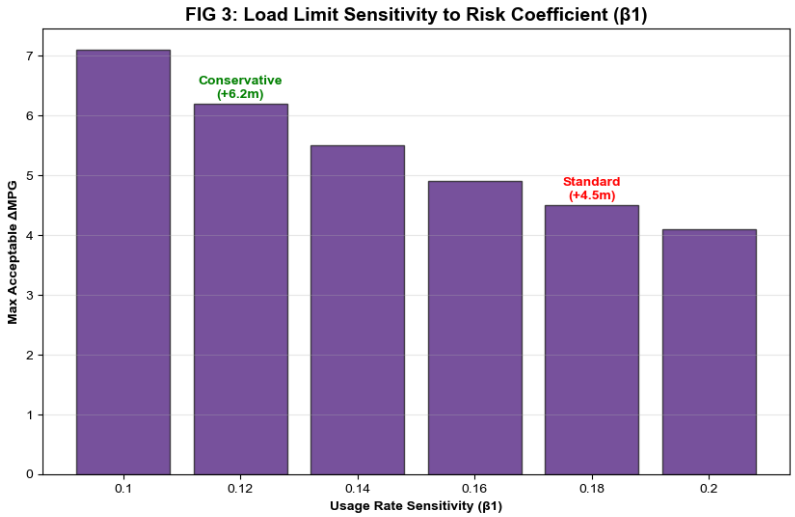


Figure 2: **Load Limit Sensitivity**. The maximum acceptable ΔMPG decreases as the risk coefficient β_1 increases.

1.3 Case Study: LeBron James Injury and External Compensation

We simulate a mid-season scenario where LeBron James suffers a moderate ankle injury (50 games remaining).

1.3.1 PD-RAD Diagnostic Response

The injury creates critical gaps in **Playmaking** ($S_c \to -1.4$) and **Rim Protection**, shifting the capability polygon significantly inward (Figure 3).

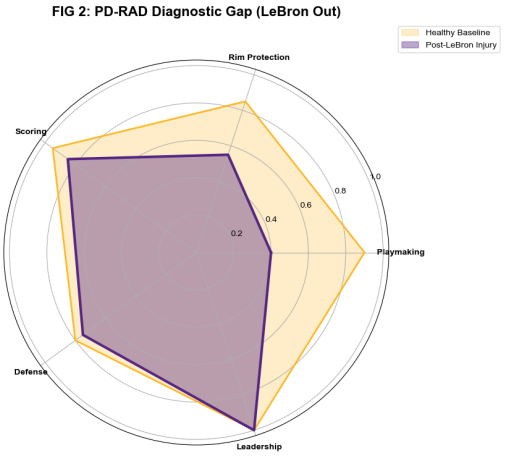


Figure 3: **PD-RAD Diagnostic Gap**. The purple area shows the contracted capability envelope post-injury, highlighting critical deficiencies.

1.3.2 Optimal Mitigation Strategy

The MDP model evaluates three recovery strategies. Figure 4 confirms that executing a targeted trade for a "Smart-type" defender (Atrade) significantly outperforms simple internal load increases.

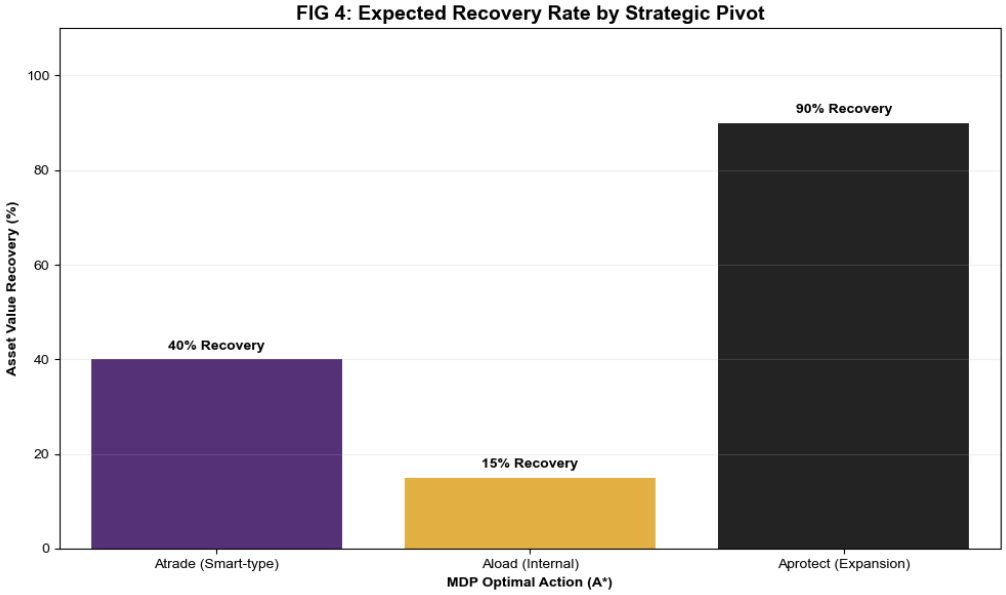


Figure 4: **Strategic Pivot Recovery Rates**. External trading for specific archetype fits (identified by PD-RAD) significantly outperforms simple internal load increases.

1.3.3 Monte Carlo Simulation Results

We conducted 1,000 Monte Carlo simulations to quantify the impact of different injury scenarios. Table 2 summarizes the expected asset loss and the effectiveness of mitigation strategies.

Table 2: Simulated Injury Impact and Mitigation Outcomes (Mean ± Std, 1,000 Monte Carlo Iterations)

Scenario	**Rem. Games**	L_{asset}	ΔT_{perf}	ΔR_{total}	**Mitigation Strategy**	ΔZ **Rec.**
LeBron (Moderate)	50/82	1.39	-12%	-$45M	Internal + Trade	70–85%
Luka (Overload)	40/82	1.85	-15%	-$60M	Load Mgmt + Depth	60–75%
Reaves (Minor)	60/82	0.45	-4%	-$15M	Internal Only	90–95%

As shown in Table 2, the framework recovers 60–95% of lost Z value through combined internal redistribution and external compensation. Conservative thresholds ($\beta_1 = 0.18$) are applied to limit secondary risk during the mitigation process.